

2.4.6

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Question

If $\mathbf{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ and $\mathbf{b} = \hat{i} + 3\hat{j} - 5\hat{k}$, show that the vectors $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$ are perpendicular.

Solution - Algebraic Approach

Two vectors are perpendicular if their dot product is zero. We will first simplify the dot product algebraically.

$$\begin{aligned}(\mathbf{a} + \mathbf{b})^T(\mathbf{a} - \mathbf{b}) &= (\mathbf{a}^T + \mathbf{b}^T)(\mathbf{a} - \mathbf{b}) \\ &= \mathbf{a}^T\mathbf{a} - \mathbf{a}^T\mathbf{b} + \mathbf{b}^T\mathbf{a} - \mathbf{b}^T\mathbf{b}\end{aligned}\tag{1}$$

Since the dot product is commutative ($\mathbf{a}^T\mathbf{b} = \mathbf{b}^T\mathbf{a}$), the expression simplifies to the difference of squares:

$$(\mathbf{a} + \mathbf{b})^T(\mathbf{a} - \mathbf{b}) = \mathbf{a}^T\mathbf{a} - \mathbf{b}^T\mathbf{b} = |\mathbf{a}|^2 - |\mathbf{b}|^2\tag{2}$$

Solution - Numerical Calculation

Now, we calculate the squared magnitudes of the given vectors:

$$\mathbf{a} = \begin{pmatrix} 5 \\ -1 \\ -3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ -5 \end{pmatrix} \quad (3)$$

The squared magnitudes are:

$$|\mathbf{a}|^2 = (5)^2 + (-1)^2 + (-3)^2 = 25 + 1 + 9 = 35 \quad (4)$$

$$|\mathbf{b}|^2 = (1)^2 + (3)^2 + (-5)^2 = 1 + 9 + 25 = 35 \quad (5)$$

Conclusion

Substituting the numerical values back into our algebraic result:

$$(\mathbf{a} + \mathbf{b})^T (\mathbf{a} - \mathbf{b}) = |\mathbf{a}|^2 - |\mathbf{b}|^2 = 35 - 35 = 0 \quad (6)$$

Since the dot product is zero, the vectors $(\mathbf{a} + \mathbf{b})$ and $(\mathbf{a} - \mathbf{b})$ are perpendicular. Hence, proved.

C Code - Calculation Function

```
// File: vector_ops.c
float check_perpendicularity(
    float ax, float ay, float az,
    float bx, float by, float bz) {

    // Calculate p = a + b
    float px = ax + bx;
    float py = ay + by;
    float pz = az + bz;

    // Calculate q = a - b
    float qx = ax - bx;
    float qy = ay - by;
    float qz = az - bz;

    // Return the dot product of p and q
    return (px * qx) + (py * qy) + (pz * qz);
}
```

Python Code - Setup and C Library Import

```
import numpy as np
import matplotlib.pyplot as plt
import ctypes
import os

# Load the compiled C library
# Compile with: gcc -shared -o vector_ops.so -fPIC vector_ops.c
lib_name = 'vector_ops.so'
c_lib = ctypes.CDLL(os.path.abspath(lib_name))

# Define C function argument and return types
c_lib.check_perpendicularity.argtypes = [ctypes.c_float] * 6
c_lib.check_perpendicularity.restype = ctypes.c_float
```

Python Code - Vector Definition and Calculation

```
# Define vectors
a = np.array([5, -1, -3], dtype=np.float32)
b = np.array([1, 3, -5], dtype=np.float32)
p = a + b # Vector a+b
q = a - b # Vector a-b

# Call the C function to get the dot product
dot_product = c_lib.check_perpendicularity(*a, *b)
print(f"Dot product from C: {dot_product}")
if abs(dot_product) < 1e-6:
    print("Vectors are perpendicular.")
```


Python Code - Plotting Logic

```
# Generate the 3D plot
fig = plt.figure(figsize=(10, 8))
ax = fig.add_subplot(111, projection='3d')
vectors = {
    '$\\vec{a}$': (a, 'green'),
    '$\\vec{b}$': (b, 'orange'),
    '$\\vec{a}+\\vec{b}$': (p, 'blue'),
    '$\\vec{a}-\\vec{b}$': (q, 'red')
}
origin = [0, 0, 0]

for name, (vec, color) in vectors.items():
    ax.quiver(*origin, *vec, color=color,
              arrow_length_ratio=0.1, label=name)
    ax.text(vec[0], vec[1], vec[2],
            f'({vec[0]},{vec[1]},{vec[2]})')
```

Python Code - Final Plotting Details

```
# Plotting settings
ax.set_xlabel('X Axis'); ax.set_ylabel('Y Axis')
ax.set_zlabel('Z Axis')
ax.set_title('3D Plot of Vectors')
ax.legend(); ax.grid(True)
ax.view_init(elev=20, azimuth=25) # Set view angle
plt.savefig("figs/Figure_1.png")
plt.show()
```

Generated 3D Plot

